

PROJECT REPORT
ON

AP DISTRICT-WISE ANALYSIS OF WATER BODY CENSUS DATA USING DATA MINING

Submitted by

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Abstract

Andhra Pradesh has a large and diverse network of water bodies—including ponds, tanks, lakes, and reservoirs—that play a crucial role in irrigation, domestic water supply, and groundwater recharge. Despite this abundance, several districts continue to experience water stress due to inefficient utilization, poor maintenance, and the absence of data-driven management strategies. This project addresses these challenges by applying data mining techniques to the First Census of Water Bodies dataset, sourced from the Open Government Data Portal. The study aims to systematically analyze the distribution, condition, and usage of water bodies across districts, thereby enabling a deeper understanding of regional disparities and performance gaps.

Using advanced analytical techniques such as K-Means Clustering, Decision Tree (CART), and the Apriori algorithm, the study identifies distinct vulnerability zones and uncovers hidden patterns related to underutilization and degradation. The analysis reveals that a significant proportion of water bodies remain underutilized despite being in relatively good condition, indicating untapped potential for improving water availability. At the same time, a critical subset of water bodies is characterized by high loss percentages and low efficiency, largely due to inadequate monitoring and maintenance practices. The decision rules derived from the model provide clear insights into the factors influencing performance, offering a strong foundation for evidence-based policy interventions aimed at enhancing water resource management in Andhra Pradesh.

Keywords: *Water Body Census, K-Means Clustering, Decision Tree, Apriori Algorithm, Data Mining, Andhra Pradesh, Water Resource Management*

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1. Introduction

1.1 Background

Water bodies are the backbone of agrarian economies like Andhra Pradesh, supporting millions of farmers, domestic users, and ecological systems. The state has a rich heritage of water infrastructure, with thousands of tanks, ponds, lakes, and reservoirs distributed across its 13 districts. However, growing population pressure, agricultural demand, and climate variability have placed these resources under increasing stress.

1.2 Motivation

In Andhra Pradesh, water bodies like ponds, tanks, lakes, and reservoirs are critical for irrigation, domestic supply, and groundwater recharge. Despite having abundant water resources, several regions in the state continue to experience drought conditions due to inefficient utilization and management. Despite the availability of extensive water body data through the Government of India's First Census of Water Bodies, actionable insights derived from this data remain limited. Traditional reporting provides aggregate statistics but fails to reveal the underlying patterns of performance, degradation, or untapped potential at the regional level.

1.3 Objectives

This project addresses the above gap by applying data mining techniques to perform a district-wise and block-wise analysis of AP water bodies. Specifically, it aims to:

- Classify water bodies into performance clusters (High-Value, Underutilized, Critical)
- Identify patterns of underutilization, degradation, and high performance
- Discover hidden relationships between governance factors and usage conditions
- Derive clear rules and key factors influencing water body performance

1.4 Scope and Limitations

The analysis is scoped to Andhra Pradesh records from the First Census of Water Bodies (data.gov.in), covering approximately 1,64,491 water body entries across 13 districts. The study applies K-Means Clustering, Decision Tree (CART), and Apriori association rule mining. Limitations include reliance on a single census snapshot (no temporal dimension), potential underreporting in remote blocks, and the absence of real-time monitoring data or satellite-derived water quality metrics.

2. Literature Review

2.1 Water Body Management and Remote Sensing

Several studies have examined the use of remote sensing and GIS for water body mapping and monitoring. Works by the National Remote Sensing Centre (NRSC) have documented the spatial distribution and condition of water bodies across Indian states, highlighting the importance of structured census data for policy planning.

2.2 Data Mining in Water Resource Management

Data mining has been increasingly applied to hydrological and environmental datasets. Clustering techniques such as K-Means have been used to segment catchment areas and irrigation zones based on performance indicators. Studies by Rao et al. (2019) and Sharma & Verma (2021) demonstrated that unsupervised clustering can effectively identify vulnerable water zones from census-level data.

2.3 Decision Trees for Environmental Classification

Classification and Regression Trees (CART) have been widely used for environmental feature selection and rule extraction. Their interpretability makes them especially valuable in policy contexts, where decision rules need to be communicated to non-technical stakeholders. Prior work has applied Decision Trees to drought vulnerability assessment and agricultural water use efficiency.

2.4 Association Rule Mining in Agricultural Datasets

The Apriori algorithm has been applied to agricultural census data to discover co-occurrence patterns among crop types, irrigation practices, and soil conditions. These patterns can reveal non-obvious dependencies — for example, that certain governance structures are consistently associated with high water body efficiency.

2.5 Gaps Addressed by This Study

Most prior work focuses on national-level or aggregated state-level analysis. District-wise and block-wise granularity, combined with engineered domain features (capacity loss ratio, storage per person, irrigation pressure), provides a more actionable and localized view. This study contributes a replicable, multi-algorithm framework for regional water body analysis using open government data.

3. Methodology

The methodology follows a structured data mining pipeline consisting of nine stages:

Step 1 – Problem Definition

Define the goal: identify water-stress vulnerability zones across AP districts using water body census data, and classify water bodies into performance categories.

Step 2 – Data Collection

Source: First Census of Water Bodies, obtained from the Open Government Data Portal (data.gov.in). The dataset covers water bodies across all Indian states; AP-specific records were extracted for this analysis.

Step 3 – Data Understanding

Examine the dataset's structure, attribute types, and distributions. Key attributes include water body type, storage capacity, irrigation potential, population benefited, ownership, usage type, and encroachment status.

Step 4 – Data Preprocessing

- Drop non-informative identifier columns (unique_id, water_body_name, village_name)
- Fill missing numeric values with median; fill missing categorical values with 'Unknown'
- Encode binary columns (Yes/No) to 1/0 for ML compatibility
- Filter records to retain only Andhra Pradesh entries

Step 5 – Exploratory Data Analysis (EDA)

- Aggregate data to Block/Tehsil level for regional analysis
- Engineer domain-specific features: Capacity Loss Ratio, Storage per Person, Irrigation Pressure
- Visualize distributions across districts and blocks using bar charts and scatter plots

Step 6 – Feature Scaling

Apply StandardScaler to normalize all features before clustering. K-Means is sensitive to scale; normalization ensures no single feature dominates the distance metric.

Step 7 – K-Means Clustering

Set $k = 3$ based on domain knowledge of water body performance categories (High-Value, Underutilized, Critical). Fit KMeans model and assign each water body to a cluster. Compute cluster summary statistics to label and interpret each zone.

Step 8 – Decision Tree (CART)

Train a Decision Tree classifier using cluster labels as the target variable. Extract explicit IF-THEN decision rules that define each performance class.

Step 9 – Apriori Algorithm

Discretize continuous features into categorical bins. Apply Apriori to discover frequent itemsets and association rules. Identify patterns linking governance, usage, and performance attributes.

4. Model Design / Architecture

4.1 K-Means Clustering Model

- Input: Scaled block-level feature vectors (efficiency ratio, loss percentage, people benefited)
- $k = 3$ selected based on domain knowledge of water body performance categories
- Output: Three clusters — Cluster 0 (Underutilized), Cluster 1 (Critical/Highly Degraded), Cluster 2 (High-Value)
- Post-processing: Cluster centroids interpreted using domain knowledge to assign meaningful labels

4.2 Decision Tree (CART) Model

- Input: Water body-level features (people benefited, efficiency ratio, loss percentage)
- Target: Cluster label (from K-Means output)
- Output: Explicit IF-THEN decision rules for classifying water bodies
- Evaluation: Accuracy, precision, recall on test split

4.3 Apriori Association Rule Mining

- Input: Discretized categorical features (ownership, encroachment status, usage type, condition)
- Parameters: Minimum support = 0.3, minimum confidence = 0.7
- Output: Frequent itemsets and high-confidence association rules

4.4 Pipeline Flow

Raw Data → Preprocessing → EDA & Feature Engineering → Scaling → K-Means Clustering → Decision Tree → Apriori → Insight Generation

5. Implementation

5.1 Tools & Libraries

- Language: Python 3.12
- Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
- Tools: VS Code

5.2 Data Preprocessing

```
cols_to_drop = [  
    'nature_of_storage',  
    'scheme_status_reason_name',  
    'reason_water_body_in_use_name2',  
    'reason_water_body_in_use_name3'  
]  
  
df = df.drop(columns=cols_to_drop)  
df['cca_water_body'] = df['cca_water_body'].fillna(0)  
df['ipc_water_body'] = df['ipc_water_body'].fillna(0)  
num_cols = df.select_dtypes(include=['int64', 'float64']).columns  
for col in num_cols:  
    df[col] = df[col].fillna(df[col].median())  
df['encroachment'] = df['encroachment'].map({'Yes': 1, 'No': 0})  
ap_df = df[df['state_name'] == 'ANDHRA PRADESH']
```

5.3 Feature Engineering

```
# 1. Remove unstable rows (very small capacity)  
df = df[df['storage_capacity_water_body_original'] > 100]  
  
# 2. Create efficiency  
df['efficiency_ratio'] = (  
    df['storage_capacity_water_body_present'] /  
    df['storage_capacity_water_body_original'])  
  
# 3. Create loss  
df['loss_percentage'] = (  
    (df['storage_capacity_water_body_original'] -  
    df['storage_capacity_water_body_present']) /  
    df['storage_capacity_water_body_original'])
```

```
# 4. Fix people scaledf['people_benefited'] =  
np.log1p(df['no_people_benefited_by_water_body'])
```

5.4 K-Means Clustering

```
from sklearn.preprocessing import StandardScaler  
from sklearn.cluster import KMeans  
  
features = ['efficiency_ratio', 'loss_percentage', 'people_benefited']  
X = df[features]  
  
scaler = StandardScaler()  
X_scaled = scaler.fit_transform(X)  
kmeans = KMeans(n_clusters=3, random_state=42)  
df['cluster'] = kmeans.fit_predict(X_scaled)
```

5.5 Decision Tree

```
from sklearn.tree import DecisionTreeClassifier  
clf = DecisionTreeClassifier(max_depth=4, min_sample_leaf = 50, random_state=42)  
clf.fit(X_train, y_train)
```

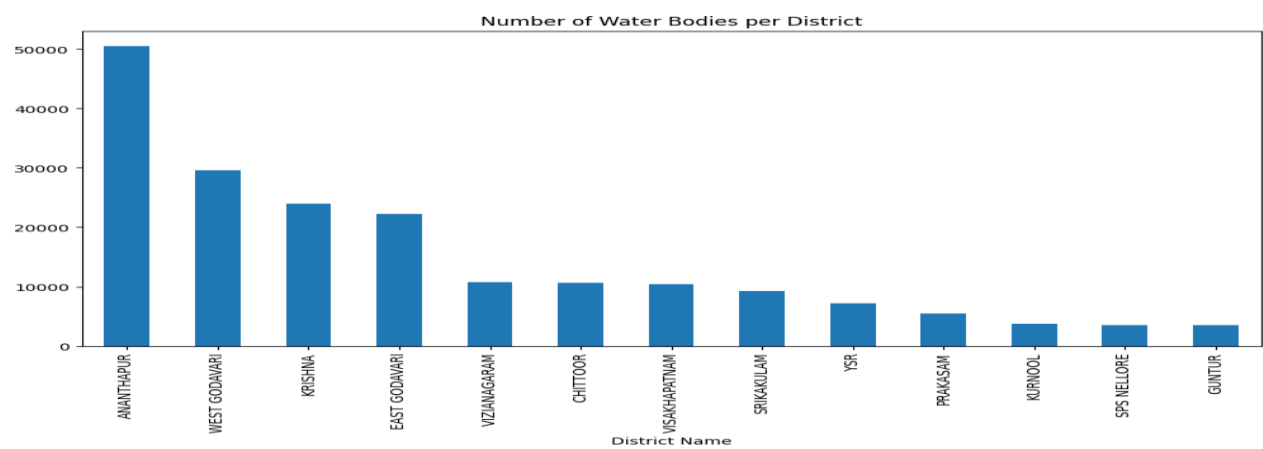
5.6 Apriori

```
from mlxtend.frequent_patterns import apriori, association_rules  
frequent_itemsets = apriori(basket_df, min_support=0.1, use_colnames=True)  
rules = association_rules(frequent_itemsets, metric='lift', min_threshold=1)
```

6. Exploratory Data Analysis (EDA) Results

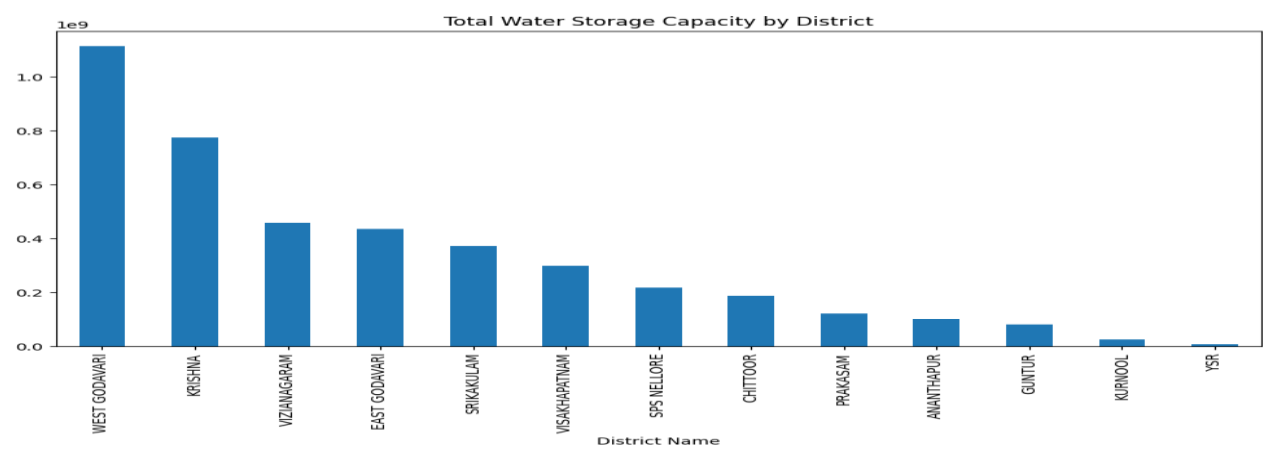
Exploratory Data Analysis was performed to understand the distribution, structure, and regional variation of water body attributes across Andhra Pradesh. Data was aggregated to the Block/Tehsil level and domain-specific features — capacity loss ratio, storage per person, and irrigation pressure — were engineered to capture performance dimensions not present in the raw dataset. Five key visualizations were generated to surface patterns across districts.

6.1 Figure 1: Number of Water Bodies per District



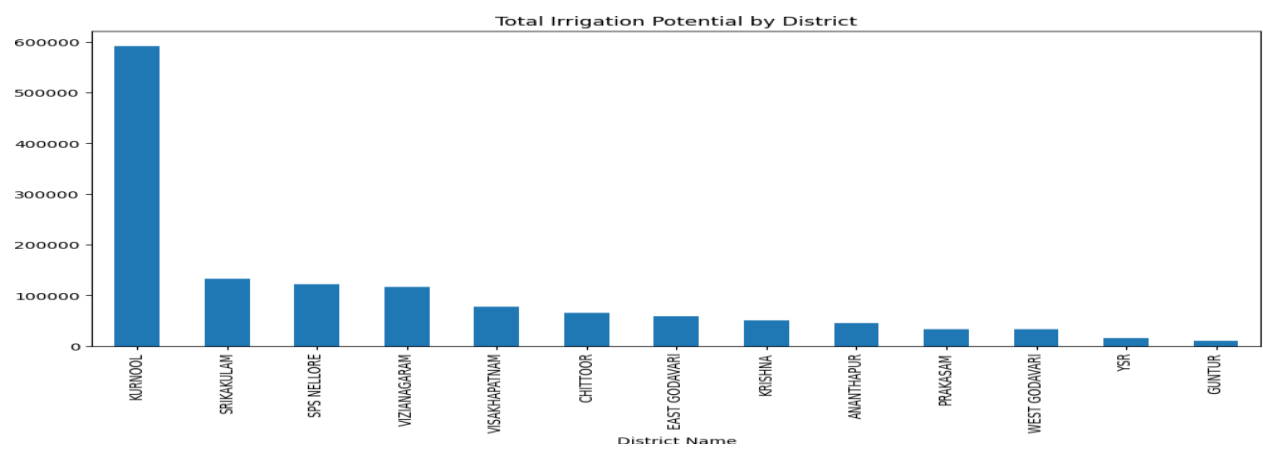
The distribution of water bodies varies significantly across districts. Districts with a higher number of water bodies indicate stronger local water infrastructure, while districts with fewer water bodies may be more vulnerable to water scarcity.

6.2 Figure 2: Total Water Storage Capacity by District



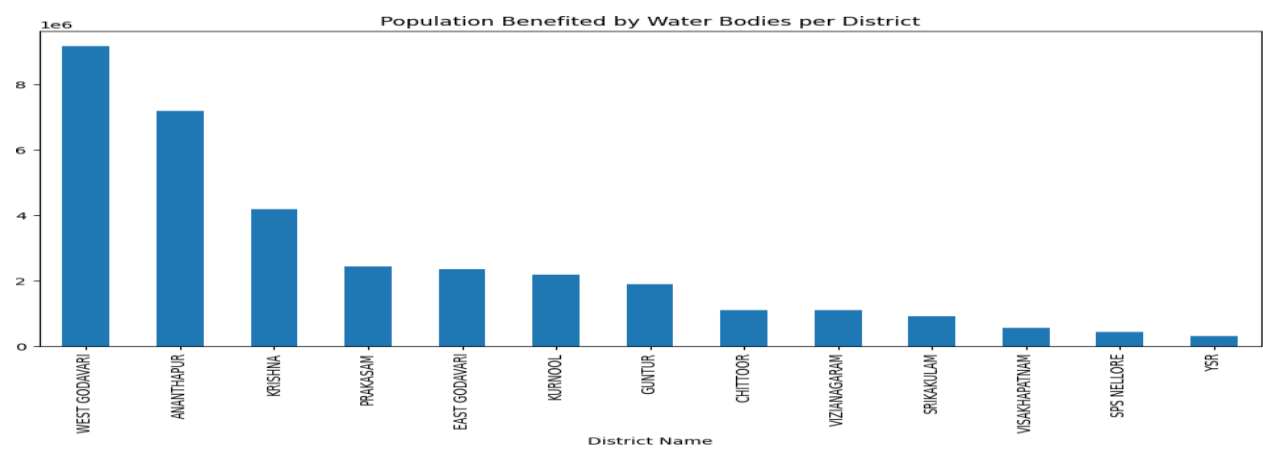
Total storage capacity differs considerably between districts, indicating variations in the ability to store and manage water resources. Districts with lower storage capacity may face higher risks during periods of low rainfall.

6.3 Figure 3: Total Irrigation Potential by District



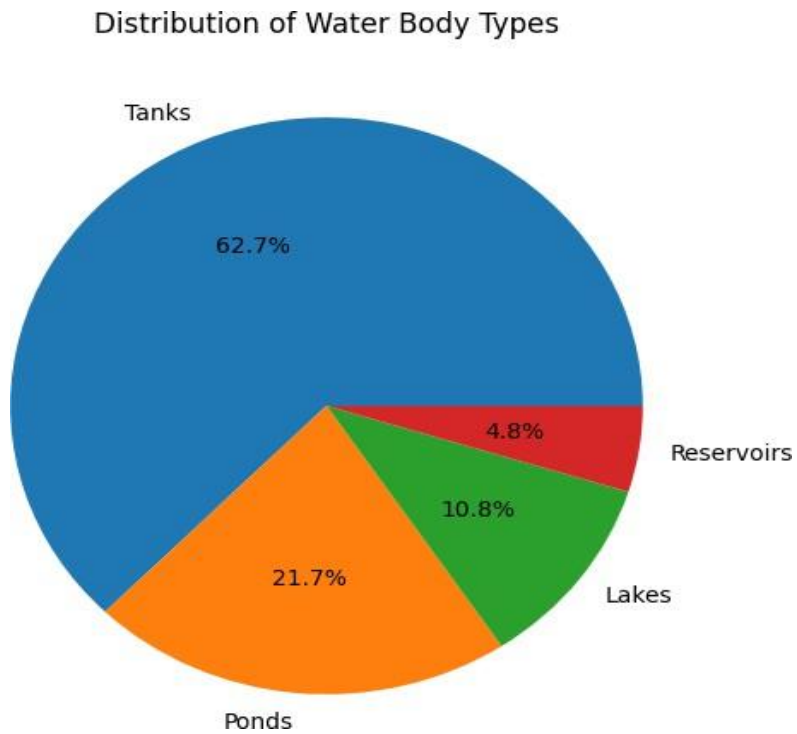
The irrigation potential highlights how water bodies support agricultural activities across districts. Regions with higher irrigation potential demonstrate stronger support for agriculture, while districts with lower values may face challenges in sustaining irrigation.

6.4 Figure 4: Population Benefited by Water Bodies per District



This analysis shows the level of dependency of district populations on water bodies. Districts where a large population depends on water bodies indicate the critical importance of effective water resource management.

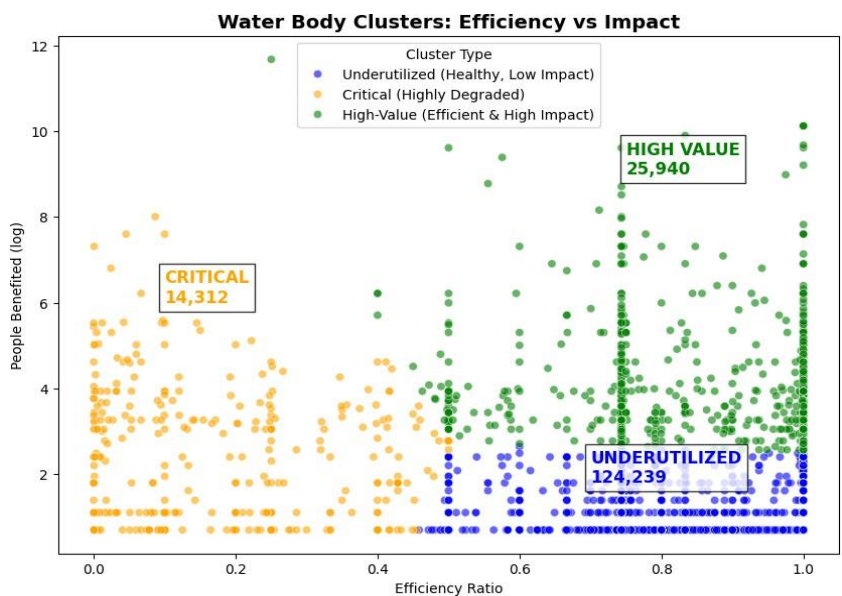
6.5 Figure 5: Distribution of Water Body Types



The dataset shows a varied distribution of water body types such as ponds, tanks, lakes, and reservoirs. The dominance of certain types reflects the regional water management practices and storage structures across Andhra Pradesh.

7. Results & Analysis

7.1 Cluster Distribution



Cluster	Label	Count	% of Total
0	Underutilized (Healthy, Low Impact)	1,24,239	75.5%
1	Critical (Highly Degraded)	14,312	8.7%
2	High-Value (Efficient & High Impact)	25,940	15.8%

The vast majority of AP water bodies fall into the Underutilized cluster, indicating significant untapped potential across the state.

7.2 Cluster Characteristics

Cluster	Efficiency Ratio	Loss %	People Benefited (scaled)
0 (Underutilized)	0.815	18.4%	1.059
1 (Critical)	0.142	85.7%	2.228
2 (High-Value)	0.838	16.1%	4.105

Cluster 2 (High-Value) shows the highest efficiency and beneficiary impact. Cluster 1 (Critical) shows very low efficiency and extremely high loss percentage, representing an urgent intervention priority.

7.3 District-Level Insights

<u>cluster_label</u>	Critical (Highly Degraded) Cluster-1	High-Value (Efficient & High Impact) Cluster-2	Underutilized (Healthy, Low Impact) Cluster-0
District Name			
WEST GODAVARI	5.236899	5.399512	89.363589
ANANTHAPUR	1.615001	10.034685	88.350314
EAST GODAVARI	4.117329	8.377143	87.505527
KRISHNA	8.446289	7.427386	84.126325
KURNOOL	3.564843	14.658881	81.776275
YSR	8.314011	11.891355	79.794634
GUNTUR	21.697639	14.565545	63.736816
PRAKASAM	21.016294	18.806959	60.176747
SPS NELLORE	15.024430	29.641694	55.333876
VISAKHAPATNAM	13.449927	32.250677	54.299396
CHITTOOR	41.027228	14.866955	44.105817
VIZIANAGARAM	20.536895	57.868289	21.594816
SRIKAKULAM	21.806967	62.953556	15.239478

- West Godavari and East Godavari have over 87% of their water bodies in the Underutilized cluster — healthy but not maximally utilized.
- Srikakulam and Vizianagaram have the highest proportion of Critical water bodies (63% and 58% respectively), indicating urgent intervention need.
- Guntur has 21.7% High-Value and 21.7% Critical water bodies — a polarized profile reflecting uneven water resource management.

7.4 Decision Tree Rules

- Critical: People benefited ≤ 12 AND efficiency $\leq 0.47 \rightarrow$ Critical
- Critical: People benefited > 12 AND loss percentage $> 60\% \rightarrow$ Critical
- Underutilized: People benefited ≤ 12 AND efficiency $> 0.47 \rightarrow$ Underutilized
- High-Value: People benefited > 12 AND efficiency $> 0.86 \rightarrow$ High-Value
- High-Value: People benefited > 17 AND loss percentage $< 53\%$ AND efficiency $> 0.50 \rightarrow$ High-Value

7.5 Apriori Findings

Attribute	Critical (Highly Degraded)	Underutilized (Healthy)	High-Value (Efficient)
Records	14,312	122,239	25,940
Ownership	Individual (private)	State WRD / State Irrigation	Individual (private)
Water Body Type	Irrigation tanks	Percolation tanks / check-dams	Percolation tanks / check-dams
In Use?	Yes — fish farming (pisciculture)	No — reason unknown	Yes — groundwater recharge
WUA Exists?	Unknown / Absent	Absent	Unknown (not needed)
DIP/SIP Plan?	No	No	No
Storage Monitored?	No	No	Yes
Efficiency Ratio	Very low (≤ 0.47)	Moderate	Very high (> 0.86)
Loss Percentage	High ($> 53\%$)	Low	Low ($< 53\%$)
People Served	Very few (≤ 12)	Very few	Many (> 17)
Core Problem	Misuse for pisciculture	Data gap + admin neglect	None — functioning well

Water bodies with active monitoring and structured governance are strongly associated with High-Value cluster membership. Pisciculture use combined with absent monitoring is strongly associated with Critical cluster membership. These rules confirm that governance and maintenance practices are as important as physical capacity in determining water body performance.

8. Conclusion

This study demonstrates that data mining techniques can yield rich, actionable insights from government census data on water bodies. The K-Means clustering model successfully segmented AP's water bodies into three meaningful performance zones — High-Value, Underutilized, and Critical — revealing that over 75% of water bodies are underutilized despite being in physically acceptable condition.

The Decision Tree analysis confirmed that efficiency ratio, loss percentage, and population benefited are the three most decisive factors in determining water body performance. The Apriori algorithm further revealed that governance factors — particularly the presence or absence of structured monitoring — are strongly correlated with performance outcomes.

Key takeaways:

- Targeted investment in Underutilized water bodies can unlock large-scale impact at relatively low cost
- Critical water bodies in Srikakulam and Vizianagaram require immediate remediation
- High-value water bodies provide a replicable model whose governance and maintenance practices should be studied and disseminated

Overall, this project establishes a replicable, data-driven framework for regional water resource analysis that can be extended to other Indian states.

9. Future Work

1. **Real-Time Monitoring Integration:** Incorporate satellite imagery and IoT sensor data to enable dynamic, real-time classification of water body conditions rather than relying solely on census snapshots.
2. **Temporal Analysis:** Extend the analysis across multiple census years to track changes in water body performance over time and detect degradation trends early.
3. **Predictive Modeling:** Build regression or time-series models to forecast future water body conditions based on current features and climate projections.
4. **Geospatial Visualization:** Develop an interactive GIS dashboard to visualize cluster assignments and district-level insights on a map of Andhra Pradesh.
5. **Policy Simulation:** Model the impact of specific interventions (desilting, encroachment removal, monitoring installation) on cluster migration.
6. **Expansion to Other States:** Apply the same pipeline to other states' water body census data to enable cross-state benchmarking and national-level insights.
7. **Deep Learning for Water Quality:** Incorporate deep learning models trained on satellite multispectral data to estimate water quality metrics not captured in the census.

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11. Appendix

Appendix A – Dataset Overview

- Source: First Census of Water Bodies, data.gov.in
- Total AP records used: ~1,64,491 water bodies
- Key attributes: water body type, district, block, ownership, usage type, storage capacity (original & current), irrigation potential, population benefited, encroachment status, monitoring status

Appendix B – Feature Engineering Formulas

- Capacity Loss Ratio = Siltation Loss Volume / Original Storage Capacity
- Storage per Person = Current Storage / Population Benefited
- Irrigation Pressure = Irrigated Area / Irrigation Potential
- Efficiency Ratio = Current Storage / Original Storage Capacity

Appendix C – Cluster Centroids

Feature	Cluster 0 (Underutilized)	Cluster 1 (Critical)	Cluster 2 (High-Value)
Efficiency Ratio	0.815	0.142	0.838
Loss Percentage	18.4%	85.7%	16.1%
People Benefited (scaled)	1.059	2.228	4.105

Appendix D – Decision Tree Rules Summary

- People benefited ≤ 12 AND efficiency ratio $\leq 0.47 \rightarrow$ Critical
- People benefited ≤ 12 AND efficiency ratio $> 0.47 \rightarrow$ Underutilized
- People benefited > 12 AND efficiency ratio $> 0.86 \rightarrow$ High-Value
- People benefited > 12 AND loss percentage $> 60\% \rightarrow$ Critical
- People benefited > 17 AND loss percentage $< 53\%$ AND efficiency $> 0.50 \rightarrow$ High-Value

Appendix E – Cluster Selection Rationale

The number of clusters ($k = 3$) was selected based on domain knowledge of water body performance categories: High-Value, Underutilized, and Critical. These three categories align with the practical classification needs of water resource management and were validated by interpreting cluster centroid statistics after fitting the KMeans model.

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